

GCSE Mathematics: Paper 3 (Calculator)

Exam Date: Wednesday, 10 June 2026

The Ultimate Topic Prediction Hit-List

Important: This hit-list is generated by analyzing topics that have already appeared on Paper 1 and Paper 2. While these topics have a very high statistical probability of appearing on Paper 3, you should still ensure you have a broad understanding of the entire syllabus.

After a rigorous analysis of Paper 1 and Paper 2, our mathematics department has identified major topics that Edexcel has **not yet tested**. Because these topics are core to the syllabus, their absence in the first two papers means they are incredibly likely to appear in Paper 3 (Calculator).

Tier 1: Extremely High Probability (Must Know)

These topics almost always appear across the three papers. Since they haven't shown up yet, prioritize these tonight:

- **Trigonometry (3D and Advanced):** Sine rule, Cosine rule, and area of a triangle ($\frac{1}{2}ab \sin C$). Expect a 4-5 mark applied question.
- **Advanced Statistics & Histograms:** Estimating the mean from grouped frequency tables, drawing or interpreting histograms (remember: frequency density = frequency $\div$ class width).
- **Quadratic Inequalities & Simultaneous Equations:** Solving simultaneous equations where one is linear and one is quadratic.
- **Vectors:** Vector proofs and geometry. Usually the last or second-to-last question on the paper.

Tier 2: Medium-High Probability (Calculator Heavy)

These topics rely heavily on calculator use and are perfect for Paper 3:

- **Compound Interest & Depreciation:** Using multipliers (e.g., 1.05^n for 5% increase over n years). Look out for questions asking you to find the number of years.
- **Bounds and Error Intervals:** Calculating upper and lower bounds in complex calculations (e.g., maximum speed = upper bound distance $\div$ lower bound time).
- **Iteration:** Using iterative formulas to find approximate solutions to equations. This is guaranteed easy marks if you know how to use the 'ANS' button on your calculator!
- **Circle Theorems:** Often combined with trigonometry or algebra. Make sure you can state the reason clearly (e.g., "Alternate segment theorem").

Top 3 Examiner Tricks to Watch Out For

- **Units mismatch:** They will give you dimensions in cm but ask for volume in litres. (Remember: $1000\text{cm}^3 = 1 \text{ Litre}$).
- **Rounding too early:** Only round your final answer. Keep the full number on your calculator display during intermediate steps, otherwise you will lose the final accuracy mark.
- **"Show that..." questions:** You must write out every single line of working. You cannot jump steps, even if you do it in your head or on the calculator.

Need to secure your A-Level grades next year?

Don't leave your A-Level success to chance. StudyHours provides elite 1-on-1 tutoring focused purely on mastering the mark scheme and securing top university placements.

Visit www.studyhours.com to book a free demo session today.

Produced by the StudyHours Mathematics Department. Not affiliated with Pearson Edexcel.